



Compressive capacity of concrete dry joints across shotcrete, extrusion, and particle-bed additive manufacturing with robotic subtractive finishing

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Abstract Additively manufactured (AM) concrete segments often require reliable, repeatable dry-joint interfaces, yet segmentation quality and dimensional accuracy remain challenging. This study evaluates dry joints produced by robotic subtractive finishing (sawing/milling) as a practical interface strategy for segmented AM concrete construction. Four joint geometries (Smooth, Triangular, Arc, and Trun-Pyramid) were fabricated as interlocking pairs from blanks produced by three AM routes (shotcrete, extrusion, and particle-bed printing). Specimens with build orientations of 0° and/or 90° were tested under monotonic uniaxial compression. Cross-method comparability was supported by batch-matched companion-cube tests, laser-scan-to-CAD geometric assessments, and process measurements, including recorded machining durations for milled/sawn configurations. In total, 120 tests were performed (102 monolithic/jointed compression tests and 18 companion-cube tests). Several joint configurations reached compressive capacities close to the corresponding monolithic references. The Arc geometry showed the most consistent high performance across all AM routes. For shotcrete printing, the sawn Triangular joint provided the most favorable performance-to-machining-time

trade-off, indicating substantial reductions in finishing effort. Joint geometry, AM route, build orientation, and subtractive finishing jointly govern both compressive capacity and fabrication efficiency. The reported dataset provides quantitative guidance for selecting dry-joint details in segmented AM concrete elements.

Keywords Concrete additive manufacturing · Dry joints · Robotic subtractive finishing · Shotcrete · Extrusion · Particle-bed printing

1 Introduction

1.1 Background: robotic prefabrication and concrete additive manufacturing

Robotic prefabrication is increasingly pursued in construction to improve repeatability, quality control, labour efficiency, and digital integration. Within this transition, concrete additive manufacturing (AM) has attracted particular interest because it can reduce formwork, accelerate prefabrication, and enable geometries that are difficult to realise by conventional casting [1, 2]. These advantages make AM attractive for bespoke components, modular systems, and digitally coordinated production chains.

For structural applications, however, concrete AM is not governed by material strength alone. Its practical viability also depends on process stability,

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dimensional repeatability, geometric accuracy, and the ability to manufacture *assembly-critical interfaces* with controlled tolerances [3]. The transition from printable elements to reliable assembled systems therefore requires suitable materials, stable deposition processes, and reproducible load-bearing contact zones.

Concrete AM comprises several process families with different implications for surface condition, geometric resolution, and structural behaviour. Extrusion-based printing is the most established route at construction scale, but layerwise deposition introduces direction-dependent behaviour, interface sensitivity, and geometric irregularities [4]. Particle-bed approaches, including binder jetting and selective activation, offer different opportunities in local resolution and shape complexity, but also involve limitations related to porosity, material processing, and scale-up [5, 6]. Across these routes, surface roughness, limited local precision, and reinforcement integration become especially critical when printed components are assembled into load-bearing systems [2, 4].

1.2 Segmented construction and the role of dry joints

Although monolithic printing of full-scale structural members is often presented as a long-term objective, practical constraints frequently favour *segmentation* followed by *assembly*. These constraints include transport limits, equipment envelope, printing stability, reinforcement placement, post-processing, and possible replacement or disassembly. In segmented construction, the joint frequently governs structural performance by controlling stress transfer, contact engagement, slip restraint, and robustness under compression, shear, and combined actions.

Dry joints are attractive in this context because they simplify erection, avoid wet-site operations, and can support disassembly, reuse, and reconfiguration [7]. Compared with wet or steel-dependent connection strategies, dry concrete–concrete interfaces offer a fabrication-compatible and potentially demountable solution, provided that load transfer is achieved reliably through geometry, contact, and confinement. At the same time, dry assembly shifts much of the structural sensitivity to the interface itself: small deviations can reduce the effective bearing area, shift the contact patch, and amplify local stress concentrations,

increasing scatter in stiffness, strength, and failure mode.

1.3 State of the art: interlocking joints, interface mechanics, and hybrid finishing

To improve the stability and reliability of dry interfaces, interlocking geometries inspired by carpentry, keyed joints, and segmental construction have been investigated for cementitious systems. Their mechanical purpose is to guide the force path toward compression-dominant transfer, mobilise friction through normal pressure, and provide geometric restraint against relative displacement. In conventional precast segmental concrete, work on dry and epoxied keyed joints has shown that joint behaviour is governed by bearing, friction, local crushing, sliding, and confinement, with peak response being sensitive to contact condition and damage near corners and key flanks [8, 9].

For additively manufactured elements, the problem becomes more complex. Process-induced anisotropy, layer topology, porosity, edge chipping, and route-specific surface defects can alter local stress distributions and failure initiation at the interface. Recent work on dry joints under combined shear and compression in shotcrete-, extrusion-, and particle-bed-based concrete has shown that interface geometry and axial precompression are dominant parameters, while AM route, build orientation, and interface preparation quality further modify the response [7]. Dry-joint performance is therefore a coupled mechanics–manufacturing problem rather than a material problem alone.

In high-performance cementitious systems, including ultra-high-performance concrete (UHPC), joints can retain high stiffness and substantial mechanical capacity, but their response remains dependent on geometry, confinement, fibre characteristics, and boundary conditions [10]. Previous studies on interlocking dry connections, CNC-fabricated interfaces, and segmented AM concrete show that geometry selection must consider strength, assembly behaviour, stress concentration, manufacturability, and tolerance sensitivity [11–14]. Under compression-dominant loading, smooth interfaces rely mainly on bearing and friction, whereas sharper or faceted profiles can introduce wedge effects, tensile splitting, or notch-controlled damage if contact is imperfect.



For dry structural interfaces, geometric fidelity is therefore a central design requirement. Reliable dry assembly demands local flatness, repeatable alignment, and predictable contact engagement. Accordingly, scan-to-CAD quality assessment and benchmark-based evaluation of geometric deviations have become important tools for quantifying accuracy, repeatability, and tolerance compliance in concrete AM.

In practice, AM alone often does not provide the surface quality and dimensional precision required for assembly-grade joints. A robust route is *hybrid additive–subtractive manufacturing*, in which near-net-shape printed blanks are post-processed by robotic sawing or milling. Such workflows combine the geometric freedom of AM with the local precision of subtractive finishing and are increasingly adopted for dry interlocking interfaces and segmental systems [7, 15]. For load-bearing joints, improved fit-up and reduced local deviation directly affect contact stability, stress redistribution, and resistance to premature local crushing.

A further manufacturing aspect is that precision is process- and time-dependent. Recent work on stamping and green-state CNC milling of dry concrete connections has shown that achievable accuracy, tool abrasion, and energy demand depend on material state, processing window, and engagement mode [16]. These findings underline that manufacturability cannot be separated from structural design when dry AM joints are considered.

Experimental and numerical approaches have both been used to study such interfaces. Compression, push-off, and shear-dominant tests help identify local crushing, splitting, interface slip, and corner damage [8]. Finite-element (FE) modelling has also been used to visualise stress redistribution, likely damage zones, and geometry effects under idealised conditions [9]. This combined strategy has been adopted in recent work on dry AM concrete joints [7, 11–14]. However, FE remains limited unless scan-based defects, anisotropy, and interface variability are explicitly resolved, making experimental evidence essential for comparative assessment across fabrication routes.

1.4 Current limitations and research gap

Despite this progress, important limitations remain. Many studies focus on a single AM route, interface

concept, or manufacturing strategy, restricting direct comparison between deposition-based and bed-based processes. Existing work also often addresses either fabrication feasibility or structural response, without explicitly linking route-dependent production characteristics to measured interface accuracy and joint performance.

In addition, while recent studies on AM dry joints under shear–compression have highlighted the importance of geometry, precompression, and fabrication quality [7], fewer studies isolate the *compression-dominant* component of dry-joint behaviour across multiple AM routes, build orientations, and subtractive finishing strategies. This response is fundamental because it reveals how effectively direct contact surfaces transfer load before more complex combined-action scenarios are considered.

A further limitation is that geometric fidelity and manufacturing effort are often discussed qualitatively rather than within a unified comparative framework. Studies rarely combine route-dependent AM production, robotic subtractive finishing, scan-to-CAD assessment, and systematic compression testing of multiple joint geometries in a single experimental matrix. Thus, the relationship between fabrication route, achievable interface quality, and compression-dominant dry-joint performance remains insufficiently resolved.

1.5 Objectives and scope of the present study

The present study addresses this gap by systematically comparing dry-joint geometries fabricated from three AM routes—shotcrete, extrusion, and particle-bed printing—combined with robotic subtractive finishing under monotonic compression. It evaluates dry interlocking concrete interfaces for segmented AM construction in terms of peak mechanical response, route-dependent geometric fidelity, and fabrication effort.

The relevance of AM in this study is not limited to visually complex connection shapes. Although the simple specimens could also be produced by conventional casting, they were intentionally simplified to isolate the effects of joint geometry, AM route, build orientation, and subtractive finishing on compression-dominant dry-joint behaviour. More complex component shapes would have added variables and reduced interpretability.



At construction scale, the main value of AM lies in the formwork-free production of large, customised, and non-standardised concrete elements within a coordinated robotic workflow. In such workflows, AM can be combined with subtractive robotic post-processing to produce assembly-critical interfaces in a consistent digital chain. This is relevant for segmented and potentially prestressed prefabricated systems, in which the position, geometry, and fabrication sequence of connections must be integrated during production. Robotic fabrication also enables connection details to be adapted to structural and assembly requirements, supporting automation, force-flow-oriented detailing, and more efficient structural configurations [17].

Accordingly, the objectives are to: (i) evaluate how AM route, build orientation, and finishing method influence the compressive response of selected dry-joint geometries; (ii) relate the mechanical response to scan-to-CAD geometric fidelity; and (iii) discuss performance in relation to fabrication effort and practical interface selection for segmented AM concrete construction. The study provides a comparative basis for selecting dry-joint details that are mechanically effective, manufacturable, and geometrically reliable within robotic prefabrication workflows.

2 Materials and methods

Cube specimens ($10 \times 10 \times (\approx)10$ cm) were machined using Computer Numerical Control (CNC) tools, including saw blades and milling cutters, at the *Digital Building Fabrication Laboratory* (DBFL), TU Braunschweig. Each geometry was produced as complementary positive/negative pairs, forming jointed specimens of $10 \times 10 \times 20$ cm. Figure 2 shows the

printed-and-machined workflow; for the particle-bed route, selected joint surfaces were also evaluated in the as-printed condition, i.e., without post-processing, to assess print-to-fit assembly. The particle-bed process is shown in Fig. 1. All joint surfaces were scanned against the corresponding CAD models to quantify geometric accuracy.

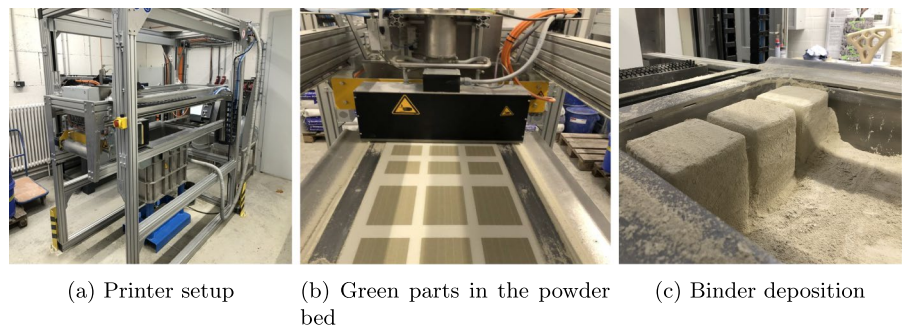
Compression tests were performed on monolithic and jointed specimens to assess the influence of joint geometry, fabrication route, and build orientation. Post-processing comprised CNC milling, sawing-based robotic finishing (SAW), and, for selected particle-bed specimens, as-printed interfaces without post-processing (Print). The specimen matrix is given in Table 1. Each tested configuration was tested with three repeats ($n = 3$), giving 102 monotonic compression tests on monolithic and jointed specimens. The orientation convention follows the jointed specimens: 0° denotes loading *parallel* to the printed layers and 90° denotes loading *perpendicular* to them. Unless stated otherwise, average results are reported.

All specimens were produced using shotcrete, extrusion, or particle-bed printing and were subsequently finished by CNC milling, sawing, or left as-printed where applicable. Detailed material and route-specific process parameters are given in Sect. 2.7.

2.1 Specimen fabrication and robotic subtractive finishing

CAD models were created in Rhino3D and processed in EasyStone as a computer-aided manufacturing (CAM) environment to generate machine-control code (G-code) for a 5-axis robotic system. Milling included roughing and finishing passes and was performed on both sides where required by tool-reach constraints. Clamping systems were calibrated to

Fig. 1 Particle-bed process: **a** printer setup, **b** green parts in the powder bed, and **c** binder deposition. Credit: iBMB TU-BS



ensure repeatable alignment (Fig. 3). Table 2 reports representative tool-path lengths and machining durations for milling and sawing, based on EasyStone simulations and practical experience.

2.2 Geometric measurement and scan-to-CAD evaluation

Dimensional accuracy of printed and machined joints was quantified by comparing 3D laser scans with CAD using a custom MATLAB pipeline

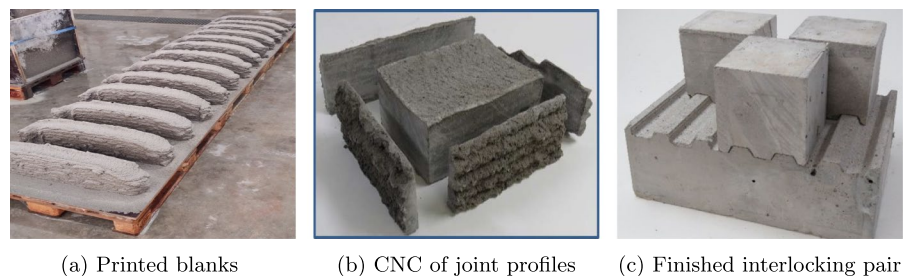
(MilledScanBig.m). The scanner acquired dense point clouds of the joint surfaces at approximately 0.1 mm resolution in X and Y in profile mode, and all data were processed in millimetres (mm). Since each connection consisted of two opposing parts, 168 joint halves (84 interlocking pairs) were scanned and compared with the nominal CAD geometries. For each joint face, the scanned window corresponded to the nominal bearing footprint of $100 \times 100 \text{ mm}^2$. The workflow included CAD-profile interpolation at 0.1 mm spacing,

Table 1 Specimen matrix of tested configurations by AM process, joint type, fabrication route, and build orientation

CNC, milling; SAW, sawing-based robotic finishing; Print, as-printed interface without post-processing. Specimen counts are aggregated over the listed orientations and fabrication routes; each tested configuration was typically repeated three times ($n = 3$). Angle notation: 0° = loading $\parallel$ layers; 90° = loading $\perp$ layers

AM process	Joint type	Fabrication(s)	Layer angle [$^\circ$]	Specimens
Shotcrete	Monolithic	–	0, 90	6
	Smooth	SAW, CNC (Milling)	0, 90	12
	Triangular	SAW, CNC (Milling)	0, 90	12
	Trun-Pyramid	CNC (Milling)	90	3
	Arc	CNC (Milling)	90	3
Extrusion	Monolithic	–	0, 90	6
	Smooth	CNC (Milling)	0, 90	6
	Triangular	SAW, CNC (Milling)	0	6
	Trun-Pyramid	CNC (Milling)	90	3
	Arc	CNC (Milling)	90	3
Particle-bed	Monolithic	–	0, 90	6
	Smooth	CNC (Milling), Print	0, 90	12
	Triangular	CNC (Milling), Print	0, 90	12
	Trun-Pyramid	CNC (Milling), Print	90	6
	Arc	CNC (Milling), Print	90	6

Fig. 2 Dry-fit concrete connection workflow: **a** printed blanks, **b** CNC milling of joint profiles, and **c** finished interlocking pairs (DBFL, TU Braunschweig; CAM: EasyStone)



(a) Printed blanks

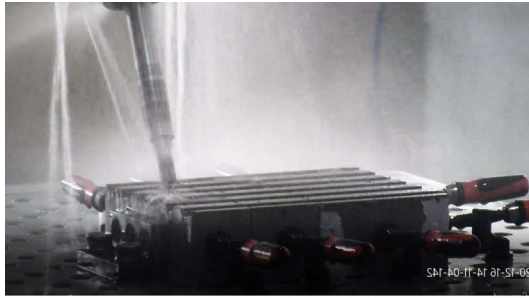
(b) CNC of joint profiles

(c) Finished interlocking pair

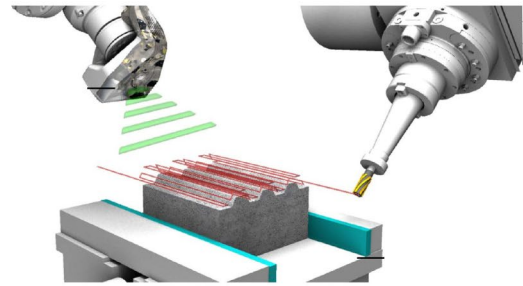
Table 2 Finishing effort for representative joints: tool-path length (Move) and machining time

Name	Smooth	Smooth	Tri-A	Tri-B	Tri-A	Tri-B	Tru-A	Tru-B	Arc-A	Arc-B
Type	Mill	Saw	Mill	Mill	Saw	Saw	Mill	Mill	Mill	Mill
Move(m)	22.05	3.2	19.44	19.85	4.8	4.6	37.36	23.60	44.59	41.09
Time(M:S)	24:15	01:05	23:34	23:43	07:29	07:13	43:15	39:40	39:53	38:13

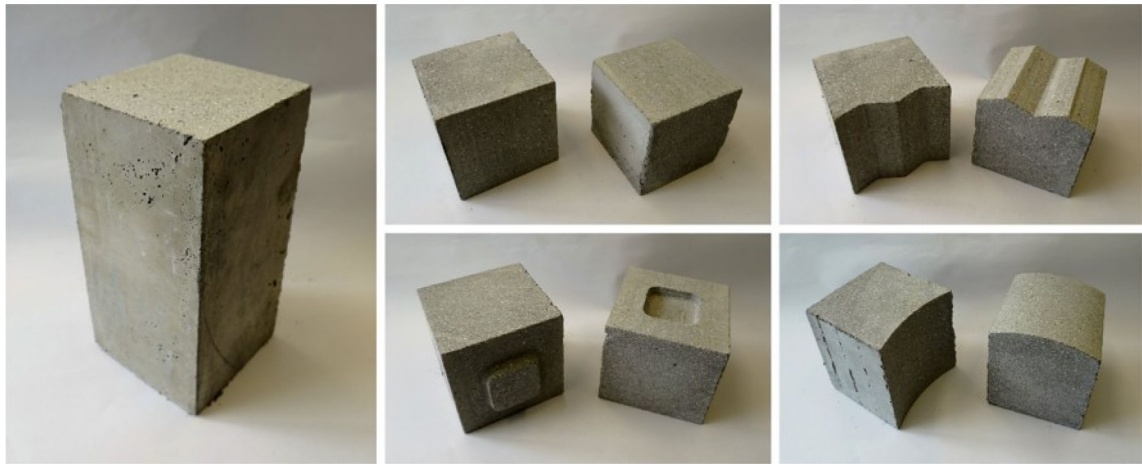
Mill, CNC milling; Tri, Triangular; Tru, Trun-Pyramid. “A” and “B” denote the two mating sides of a connection



(a) Wet cutting on the fixture



(b) CAM toolpath planning



(c) Representative manufactured specimens from the pre-test production stage (various joint profiles and finishes). Images document fabrication outputs prior to screening/testing and do not indicate damage or rejection.

Fig. 3 Manufacturing and finishing: multi-axis milling of joint profiles (top) and representative manufactured specimens from the pre-test production stage (bottom). Credit: iBMB TU-BS

scan–CAD registration by planar offsets (LOCX, LOCZ), cropping to a common domain, and point pairing to compute vertical height differences H_{error} .

After rigid registration, pointwise deviations h_i were evaluated as signed scan-to-CAD distances over the cropped interface window. The primary accuracy metric is the root-mean-square error, $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N h_i^2}$, where N is the number of sampled points. For completeness, the mean squared error is $\text{MSE} = \frac{1}{N} \sum_{i=1}^N h_i^2$, the standard deviation is $\sigma(h) = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (h_i - \bar{h})^2}$, and the maximum absolute deviation is $\text{Max} = \max_i |h_i|$. RMSE is used as the main indicator of geometric

fidelity, while Max supports defect screening and scatter interpretation. Specimens are denoted by shotcrete (S), extrusion (E), and particle-bed (P); post-processing is indicated as CNC, SAW, or Print, and build orientations are reported as 0° and 90° with respect to the loading direction.

2.3 Companion material tests

To enable direct comparison across manufacturing routes, mechanical performance is reported primarily in terms of nominal compressive stress σ_{max} (MPa), with the corresponding peak load P_{max} (kN) reported as a supplementary measure. Batch-matched companion cubes ($100 \times 100 \times 100 \text{ mm}^3$) were tested for each

AM route and build orientation to quantify the reference compressive strength f_c used for normalisation in the Results section. The companion cubes were tested under the same compression protocol described in Sect. 2.4.

2.4 Compression test setup and protocol

Compression tests were performed under displacement control at $v = 0.01$ mm/s ($= 0.6$ mm/min) until failure. Horizontal (lateral) and vertical (axial) strains were measured using two opposing strain gauges per direction and reported as gauge-averaged values. For both monolithic and jointed specimens, the nominal compressive stress at peak load was computed using the constant nominal bearing area $A_{\text{nom}} = 100 \times 100 \text{ mm}^2 = 10,000 \text{ mm}^2 = 0.01 \text{ m}^2$: $\sigma_{\text{max}} = \frac{P_{\text{max}}}{A_{\text{nom}}}$.

The gauge-averaged horizontal (transverse) strain is denoted by ε_h and defined as the arithmetic mean of the horizontal strain-gauge readings; throughout the manuscript, ε_h values are reported in %.

2.5 Strength-normalised comparison framework

To reduce bias arising from route-dependent mix designs and strength levels across the three additive-manufacturing (AM) processes, the peak response was additionally expressed using a strength-normalised joint-efficiency factor. All specimens shared the same nominal bearing footprint ($100 \times 100 \text{ mm}^2$), i.e., $A = 0.01 \text{ m}^2$. The nominal peak compressive stress was calculated as

$$\sigma_{\text{max}} = \frac{P_{\text{max}}}{A}, \quad (1)$$

where P_{max} is the measured peak load and A is the nominal bearing area. For each AM route and build orientation ($0^\circ/90^\circ$), σ_{max} was normalised by the batch-matched companion compressive strength $f_{c,\text{comp}}$, yielding the strength-normalised joint efficiency

$$\eta_\sigma = \frac{\sigma_{\text{max,joint}}}{f_{c,\text{comp}}}. \quad (2)$$

Here, η_σ represents the fraction of companion strength mobilised by a given joint configuration. In this context, a “family” denotes a fixed combination of AM route, joint geometry, post-processing

concept, and build orientation. Where η_σ remains approximately stable within a given family over a limited strength range, a practical within-family scaling may be written as

$$\sigma_{\text{max,pred}} \approx \eta_{\sigma,\text{family}} f_{c,\text{new}}. \quad (3)$$

This scaling is intended only for interpolation within the same family and specimen size, and not for transfer across different AM routes or fundamentally different interface and finishing concepts.

2.6 Finite-element modelling (qualitative)

Finite-element (FE) analyses were performed in Abaqus only as a qualitative interpretation tool for geometry-driven stress redirection, contact evolution, and likely damage-localisation zones. The model is not intended to predict peak load, nonlinear force–deformation response, or specimen-specific failure. Therefore, all quantitative capacity and deformation evaluations are based solely on the experiments, while the FE results support only the interpretation of stress and damage localisation.

For the FE analyses, Rhino-based CAD models of the dry joints were imported into Abaqus. Rigid platens were used to represent the loading setup: the bottom platen was fully constrained, while the top platen was displacement-controlled. Platen–concrete and joint-interface contacts were defined using hard normal contact and penalty-based tangential behaviour with a constant Coulomb friction coefficient of $\mu = 0.30$. This value was adopted only as an idealised representative input for qualitative comparison of geometry-driven stress-transfer mechanisms. Route-specific roughness, local surface damage, and calibrated tangential contact laws for the different interface conditions were not explicitly investigated; therefore, the FE results are not used for quantitative capacity prediction.

Concrete was modelled with the standard Concrete Damage Plasticity (CDP) formulation as an effective isotropic continuum model. The model was selected to obtain stable qualitative stress and damage fields under identical numerical assumptions. AM-specific features such as layerwise anisotropy, interlayer bond variability, route-dependent porosity, chipping, cavities, and scan-based surface deviations were not explicitly modelled. Thus, orientation effects



are evaluated experimentally, while the FE results are used only to interpret geometry-driven contact behaviour and stress-path development. The simulations assume nominal CAD geometry and idealised interfaces; defect-resolved modelling would require scan-based mapping and calibrated damage parameters, which are outside the present scope.

Model details. Concrete was meshed with 8-node linear brick elements with reduced integration (C3D8R), using a uniform characteristic element size of 10 mm for all geometries. This mesh size was adopted to maintain a consistent basis for qualitative comparison of stress-transfer and damage-localisation patterns across the different joint forms, rather than to establish a mesh-optimised predictive simulation of the experiments. To mirror the experiments, no reinforcement was included; load transfer across the joint interface was represented solely through contact and friction. CDP with post-peak softening (tension softening defined via a bilinear law) can exhibit mesh dependency when localisation develops. Therefore, the numerical results are interpreted only in terms of relative stress patterns and likely damage zones obtained under the same meshing strategy. A full mesh-convergence or fracture-energy regularisation study, for example mesh-objective softening based on tensile fracture energy, is beyond the present qualitative scope.

Initial mechanical inputs—compressive strength (f_c), tensile strength (f_t), and elastic modulus (E)—were taken from the experimental material characterisation. The remaining CDP parameters were adopted from the literature and prior DBFL modelling practice to obtain stable qualitative fields. Tensile behaviour followed a bilinear law [18, 19], compression followed Eurocode 2 (EN 1992-1-1) [20], and CDP flow/damage parameters were taken from [21]. These parameters were not calibrated to reproduce the peak

load of each individual specimen; therefore, the FE results are interpreted only as qualitative support for the experimentally observed stress-transfer and damage-localisation mechanisms.

2.7 Joint geometry and materials

Four dry-joint profiles were selected to span a representative range of contact conditions and manufacturability for the subsequent compression tests: *Smooth* (flat interface), *Triangular* (interlocking with sharp features), *Arc* (curved interlock), and *Trun-Pyramid* (tapered interlock). The selection aimed to balance (i) robotic post-processing feasibility (sawing or CNC milling), (ii) alignment and tolerance sensitivity, and (iii) expected contact-pressure uniformity under compressive loading. These four geometries were therefore used consistently throughout the experimental campaign (Fig. 4).

Although the present study focuses on compression, the comparison of joint geometries is not limited to nominal bearing area. In segmental concrete systems, especially those intended for dry assembly and possible prestressing or post-tensioning, interface geometry also governs contact-stress distribution, confinement, mechanical interlocking, and damage initiation. Even under nominally uniaxial compression, inclined or curved surfaces can redirect the load path, generate lateral force components, modify frictional engagement, and promote tensile splitting near geometric discontinuities. The tested geometries were therefore compared in terms of both compressive capacity and damage localisation, as a first step before future studies address shear, prestressing-related actions, and combined loading.

Shotcrete- and extrusion-based specimens were produced from a commercially premixed fine-grained shotcrete mortar supplied by MC-Bauchemie

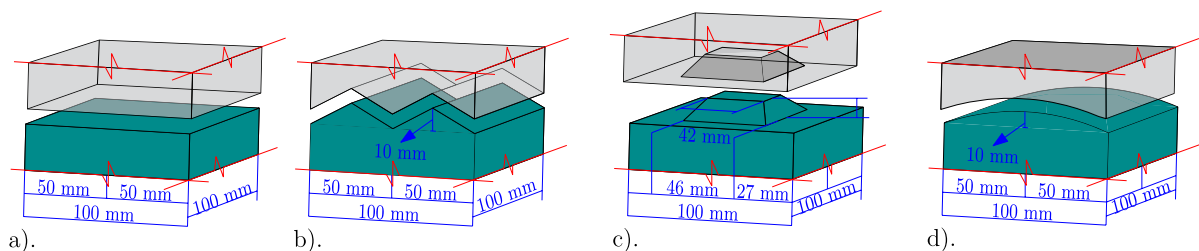


Fig. 4 Selected joint profiles and nominal dimensions: **a** Smooth, **b** Triangular, **c** Trun-Pyramid, and **d** Arc



Müller GmbH & Co. KG (Bottrop, Germany). The premix was based on Ordinary Portland Cement (CEM I 52.5 R) and, according to the manufacturer and related prior work, contained supplementary cementitious constituents (including pozzolanic additions and silica fume), fine aggregates with a maximum grain size of $d_{\max} = 2$ mm, powdered chemical admixtures, and micro polypropylene fibres. The mortar was prepared by adding water according to the manufacturer's recommendation (3.6 L per 25 kg bag) and mixed until a homogeneous, pumpable consistency was obtained. Workability was controlled using the flow-table test according to EN 12350-5.

Since the shotcrete/extrusion material was supplied as a proprietary commercial premix, the exact constituent mass fractions and full mixture proportions were not available for disclosure. To improve reproducibility despite this limitation, the supplier, cement type, maximum aggregate size, water addition, and the main route-specific process parameters are reported here.

For shotcrete printing, fabrication was carried out using a six-axis Stäubli gantry robot. A continuously pumped cementitious mix was activated at the nozzle by compressed air and a liquid accelerator. The main process parameters were a nozzle travel speed of 4500 mm/min, a concrete flow rate of 0.4 m³/h, a compressed-air flow of 30 Nm³/h, and a nozzle stand-off distance of 150 mm.

For extrusion printing, the same premixed base material was used, but without air support and without accelerator. The printhead had a rectangular outlet of 3 × 10 cm and operated with an effective standoff of less than approximately 2 mm, achieved by light skimming over the previous layer. The travel

speed was 3500 mm/min and the material flow rate was approximately 0.4 m³/h.

Particle-bed specimens were produced using Selective Cement Activation (SCA) on a commercial binder-jetting system (ZPrinter 310) with a layer thickness of 0.10 mm. The powder bed consisted of a dry blend of (i) a Portland cement containing mayenite (C₁₂A₇; “cement A”, CEM I 52.5) and (ii) quartz sand in three particle-size grades. The bed composition was kept constant at a volumetric cement-to-sand ratio of 40/60. Activation was achieved by jetting a liquid consisting of 90 vol% deionised water and 10 vol% isopropanol. The effective water input was controlled through the printer saturation setting.

3 Results and discussion

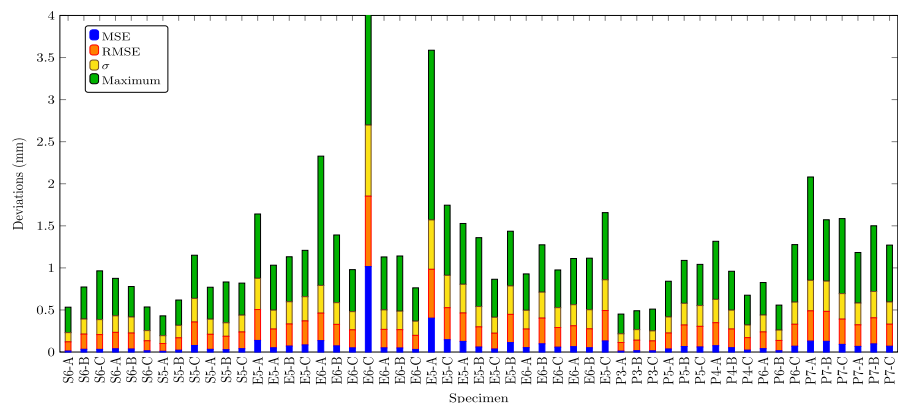
3.1 Geometrical quality assessment

Figure 5 summarises the scan-to-CAD deviation metrics for the tested joint families. Overall, shotcrete specimens generally showed the lowest deviations, extrusion specimens displayed a wider spread, and particle-bed specimens were more sensitive to geometry complexity and surface condition.

3.1.1 Shotcrete joints (S)

Shotcrete interfaces that were CNC milled generally showed the smallest global deviations, with RMSE values mostly in the sub-millimetre range. The best smooth CNC specimens reached RMSE values on the order of 0.1 mm, although some cases showed higher local mismatch. Slight 0°/90° differences indicate

Fig. 5 Scan-to-CAD deviation metrics (MSE, RMSE, $\sigma(h)$, and Max) for shotcrete (S), extrusion (E), and particle-bed (P) specimens by ID groups (A–C)



possible effects of surface formation and clamping during finishing. Overall, low interface misfit in shotcrete requires consistent post-processing and careful scan registration. The full deviation metrics are shown in Fig. 5.

3.1.2 Extruded joints (*E*)

Extruded specimens displayed a wider spread of geometrical deviations compared with shotcrete, reflecting the combined influence of layer-wise deposition, time-dependent material stability, and tool accessibility during finishing. For triangular profiles, both SAW and CNC post-processing achieved sub-millimetre RMSE values for well-controlled specimens; however, deeper or less accessible contours were more prone to increased local deviations. This behaviour is consistent with sensitivity to fixture compliance/vibration during milling and with limited tool reach on pronounced interlock geometries. The corresponding distributions of RMSE and *Max* values are included in Fig. 5.

3.1.3 Particle-bed geometries (*P*)

For particle-bed printing, post-processing by CNC milling markedly improved geometric fidelity relative to as-printed interfaces. Smooth, CNC-finished specimens typically achieved low RMSE values comparable to the best shotcrete cases, whereas as-printed interfaces showed larger deviations and stronger orientation sensitivity. More complex geometries (e.g., Trun-Pyramid) exhibited the largest deviations and the highest variability, indicating that geometric complexity and local fragility dominate achievable accuracy in this workflow. Figure 5 summarises the accuracy metrics across the particle-bed series.

3.1.4 Observed defects and practical implications

Beyond global dimensional deviations, local surface defects were observed on a subset of specimens, including edge chipping, spalling, voids, handling abrasion, and milling-induced edge breakage (Fig. 6). Such defects can reduce the effective contact area and promote local stress concentrations, thereby contributing to scatter in peak capacity and to geometry-dependent crack initiation. In summary, shotcrete with CNC finishing provides the most consistent geometric fidelity for smooth interfaces, extrusion exhibits a broader accuracy spread influenced by feature depth and post-processing accessibility, and particle-bed printing benefits strongly from CNC finishing but remains highly geometry- and orientation-sensitive for complex interlock profiles.

3.2 Finite-element interpretation (qualitative)

As stated in the Methods section, the FE model is retained only as a qualitative interpretation tool and is not presented as a predictive representation of the experiments. AM-specific features such as layerwise anisotropy, interlayer imperfections, chipping, voids, scan-based surface deviations, and route-specific tangential contact behaviour were not explicitly modelled. Accordingly, the FE results are used only to visualise stress redistribution, contact engagement, and likely damage-localisation patterns under idealised conditions. The experimental results remain the sole basis for the quantitative evaluation of load capacity and deformation response (Fig. 7).

3.2.1 Monolithic reference specimen

Figure 8 juxtaposes the experiment and the FE simulation for a monolithic shotcrete specimen

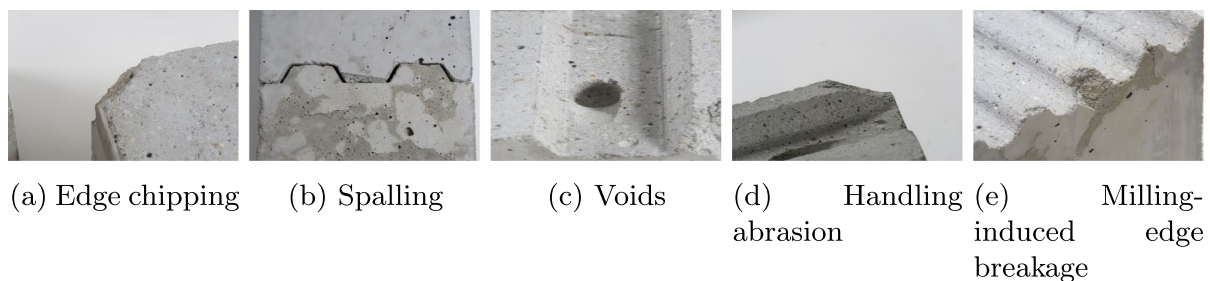


Fig. 6 Typical surface features observed on printed and milled joints



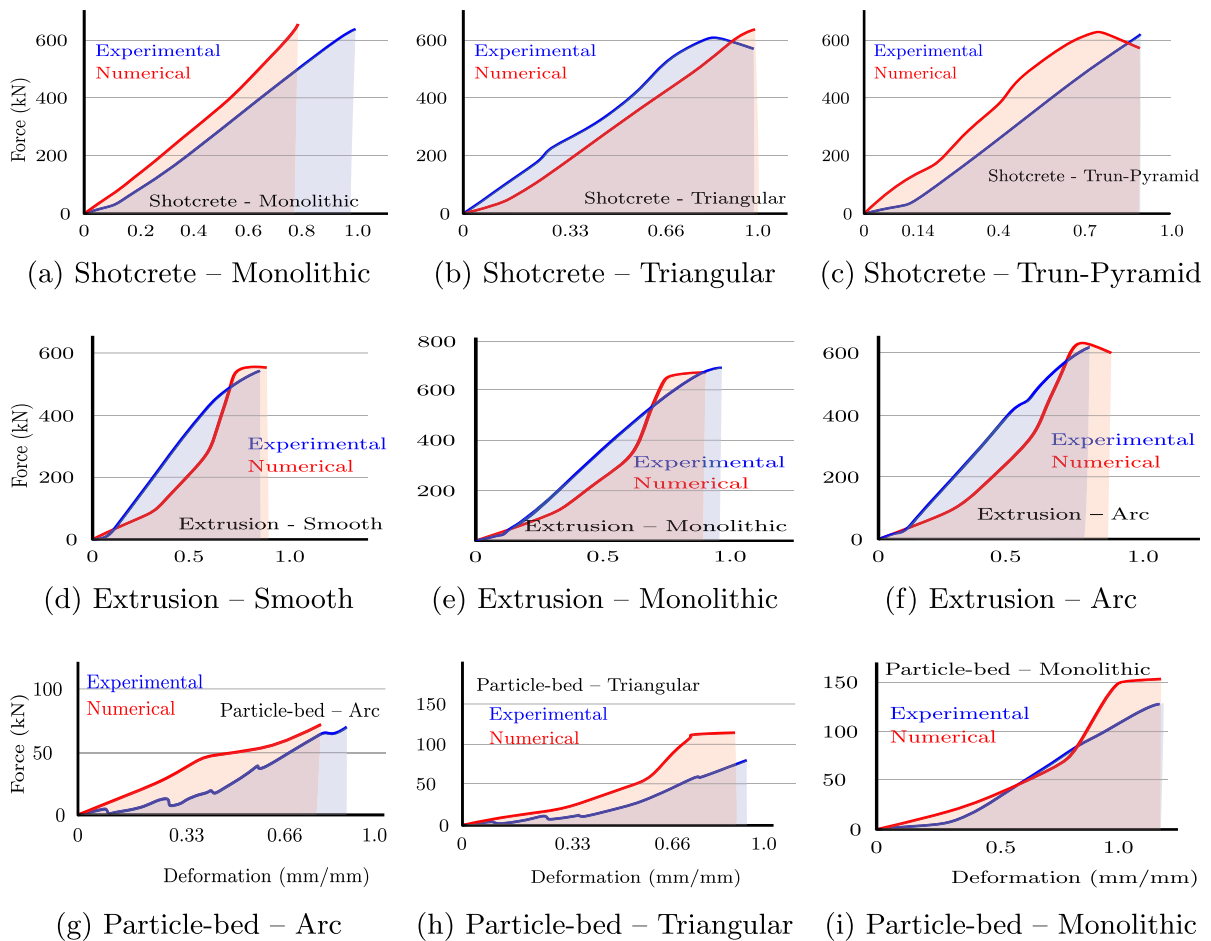


Fig. 7 Representative experimental force–deformation diagrams shown together with FE responses used only as qualitative interpretation aids for geometry-dependent stress-transfer

trends. The FE results are not used for quantitative validation of peak load, stiffness, or deformation capacity.

under uniaxial compression, highlighting qualitative similarities in stress redistribution and damage localisation.

The experimental image shows failure initiating as vertical splitting along the load axis, followed by corner spalling and short inclined branches, consistent with the generally brittle response of unreinforced concrete under platen friction.

Although cone- or inverted-pyramid-type crushing may occur in compression tests due to platen confinement, vertical splitting with local corner spalling is also a common failure mode for brittle, unreinforced concrete specimens. This response can be promoted by platen friction, local surface irregularities, material heterogeneity, and stress

concentrations. Therefore, the photograph is interpreted together with the measured force–deformation response and repeated tests, and is not considered evidence of a testing error.

The FE contours (S22 and DAMAGEC) show qualitatively compatible features, including a compressed core, reduced compression toward free surfaces (barreling), and localized compressive damage along the sides and near the top edges. These fields are consistent with the observed tendency for splitting and corner spalling, but they are not used as a quantitative validation of the failure process.

Longitudinal section cuts show a compressed core with reduced compression near the free surfaces, consistent with barreling under platen

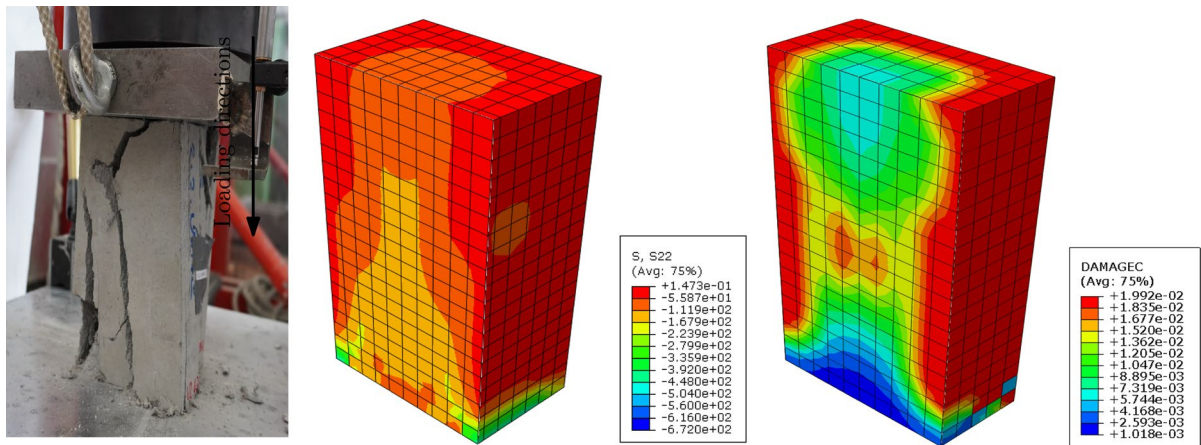


Fig. 8 Monolithic specimen (shotcrete): compression test (left, *Credit: iBMB TU-BS*) and qualitative FE contour plots from longitudinal sections (middle/right). The arrow indicates the loading direction; S_{22} is reported along the loading axis and

$DAMAGEC$ denotes compressive damage. The FE contours are used only to interpret stress and damage localisation patterns, not for quantitative validation of capacity or deformation

restraint. The $DAMAGEC$ field localizes along the sides and near the upper edges, qualitatively matching the observed splitting and corner spalling. These fields are used only to illustrate likely stress redistribution and damage localisation.

3.2.2 Smooth joints

Figure 9 compares the experiment with the FE simulation for Smooth shotcrete joints under uniaxial

compression. The FE results are used only for qualitative interpretation of contact engagement and likely damage-localisation trends. Cracking initiates at or near the joint plane—often in the upper third—and propagates along the smooth transition where mechanical interlock is minimal; localized crushing develops near the platen contacts, and short vertical splits form in the adjacent legs. The FE contours show stress concentration along the flat mating surfaces and at the top/bottom interfaces, while the

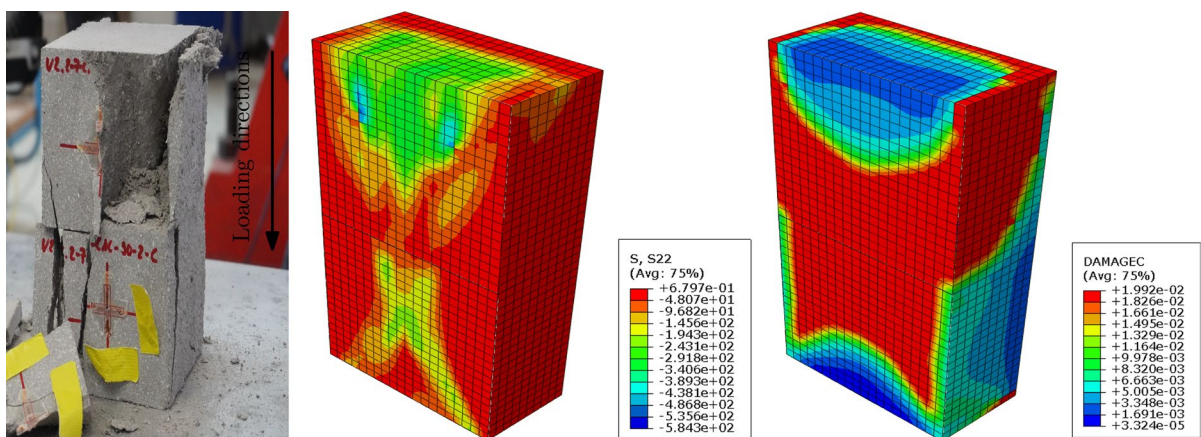


Fig. 9 Smooth joint (shotcrete): compression test (left, *Credit: iBMB TU-BS*) and qualitative FE contour plots from longitudinal sections (middle/right). The arrow indicates the loading direction; S_{22} is reported along the loading axis and $DAMAGEC$

shows compressive damage. The FE contours are interpreted qualitatively and are not used for quantitative validation of the experimental response

DAMAGEC field localizes on the joint surface and near the upper corners. Although the FE model does not reproduce all AM-specific imperfections, these qualitative trends are consistent with the experimental observation that the Smooth interface provides limited mechanical interlock and remains sensitive to local contact conditions.

3.2.3 Analysis of triangular joints

Figure 10 combines experimental observations and FE results for Triangular joints produced by shotcrete printing under uniaxial compression. The FE results are used only to visualise likely stress-transfer and damage-localisation trends under idealised conditions. The tests show diagonal shear-splitting cracks that nucleate at the inclined interfaces and coalesce, leading to fragmentation and toppling of the two halves. The geometry redirects part of the axial load into shear and lateral components, so failure is interface-dominated rather than purely compressive.

The FE fields indicate strong compression in the lower central core and along narrow vertical strips, while the stress trajectories rotate around the Triangular cut-outs. The compressive-damage map (DAMAGEC) localizes near the ridge tips and along the oblique flanks of the Triangular recesses, which is qualitatively consistent with the observed crack paths and corner losses. These results suggest that the notch geometry strongly

guides the force path and promotes early localization near the ridge tips and inclined flanks.

3.2.4 Analysis of arc joints

Figure 11 combines experimental observations and FE results for Arc joints produced by shotcrete printing and loaded in uniaxial compression. The FE fields are included only for qualitative interpretation of geometry-dependent stress transfer and crack initiation. In the tests, cracks first appear at the shoulders of the Arc where curvature changes and normal pressure peaks. Splitting then develops along the vertical legs, while short inclined branches grow from the intrados toward the loading platen. Compared with Smooth joints, the curved interface guides the load into compressive thrust along the Arc, so damage spreads over a broader band and the onset of global splitting is delayed.

The FE fields show high compression at the Arc shoulders and at the base supports, with stress rotation around the curved interface and partial unloading at the intrados. The tensile-damage contour (DAMAGET) localizes above the crown and along the flanks, which qualitatively follows the experimentally observed crack pattern. These contours suggest that the curved geometry promotes a broader and more continuous load-transfer path than the Smooth joint.

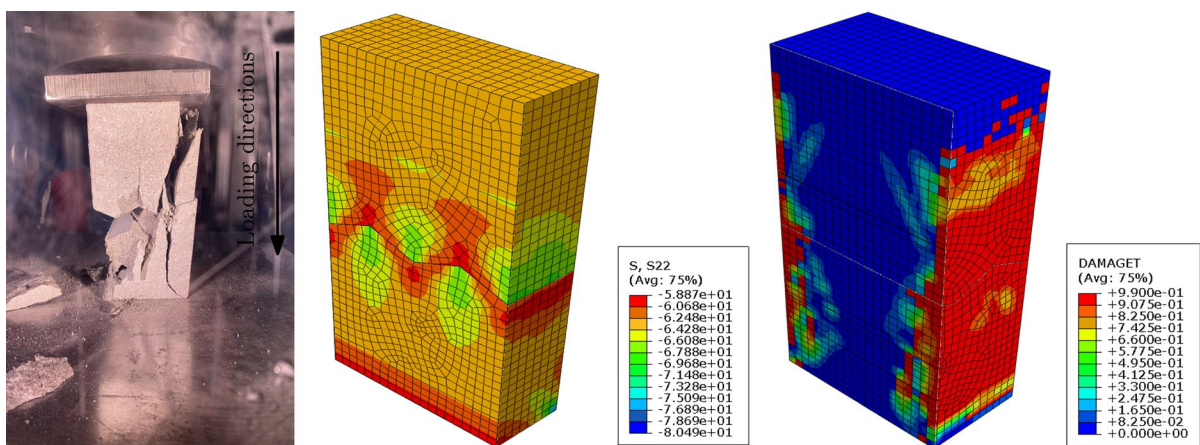


Fig. 10 Triangular joint (shotcrete): compression test (left, Credit: iBMB TU-BS) and qualitative FE contour plots from longitudinal sections (middle/right). The arrow indicates the loading direction; S_{22} is reported along the loading axis and

DAMAGEC shows compressive damage. The FE contours are used only to visualise likely stress-transfer and damage-localisation trends under idealised conditions

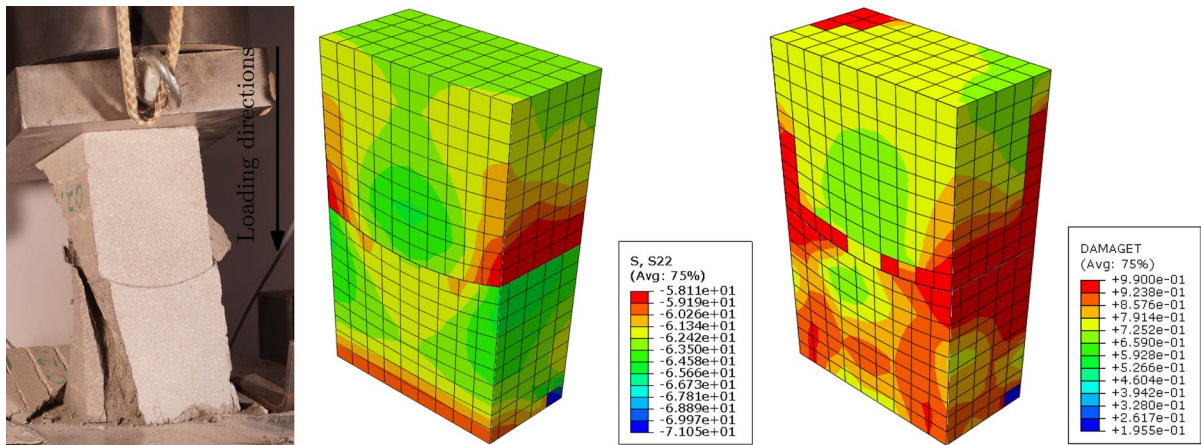


Fig. 11 Arc joint (shotcrete): compression test (left, *Credit: iBMB TU-BS*) and qualitative FE contour plots from longitudinal sections (middle/right). The arrow indicates the loading direction; S_{22} is reported along the loading axis and $DAMAGE$

shows tensile damage. The FE contours are used only for qualitative interpretation of geometry-dependent stress redistribution and crack-initiation tendencies

In the Arc joint, the curved geometry redirects the thrust and reduces net sliding demand, so failure tends to initiate at the shoulders where curvature and contact pressure change.

3.2.5 Analysis of Trun-Pyramid joints

Figure 12 combines experimental observations and FE results for the Trun-Pyramid joint. The numerical fields are used only to interpret qualitative stress

concentration and damage-localisation patterns under idealised conditions.

In the tests, cracking nucleates at the narrowed throat of the recess and at the lower notch corners, then spreads upward along the sloped flanks and downward toward the base, leading to fragmentation of the upper block.

The FE fields show peak compression in the throat of the Trun-Pyramid and at the shoulders where the slope changes. At the same time, the outer faces

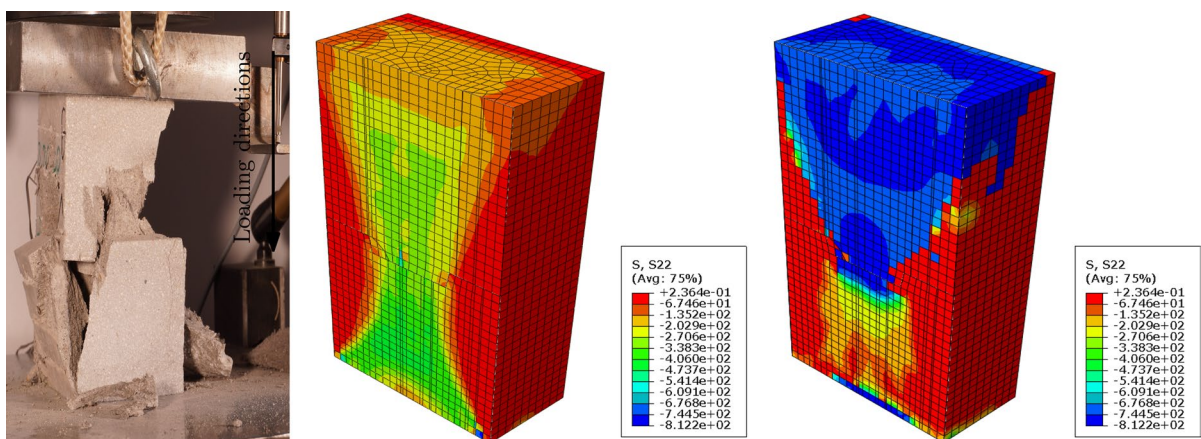


Fig. 12 Trun-Pyramid joint (shotcrete): compression test (left, *Credit: iBMB TU-BS*) and qualitative FE contour plots from longitudinal sections (middle/right). The arrow indicates the loading direction; S_{22} is reported along the loading axis and

$DAMAGE$ shows tensile damage. The FE contours are interpreted only as qualitative indicators of stress concentration and damage-localisation patterns

experience lower and more uniform compression, indicating limited lateral confinement. The tensile-damage map (DAMAGET) localizes in the lower central region and along the oblique flanks, consistent with the observed diagonal cracking. In practice, any imperfect surface conformity between the mating faces may further intensify this behaviour by reducing the true contact area and introducing a wedge-like load-transfer effect. This can generate local tensile stresses perpendicular to the inclined faces and thereby promote earlier splitting. The Trun-Pyramid geometry is especially sensitive to this mechanism because the inclined surfaces and narrowed throat already concentrate the compressive force path into a limited contact region.

Thus, friction and local normal pressure contribute to shear transfer, but crack initiation is governed mainly by stress concentration at the throat and slope transitions.

Comparison of failure mechanisms Across the tested joint families, the failure patterns indicate how the interface geometry redirects compressive load paths and concentrates local confinement and shear tractions. Smooth interfaces provide little mechanical interlock, leading to joint-plane-dominated localisation. Arc joints redirect thrust through the curved shoulders, promoting more distributed engagement and delaying global splitting. Notch-type geometries (Triangular and Trun-Pyramid) intensify stress localisation near geometric discontinuities: Triangular joints concentrate demand near ridge tips and oblique flanks, while Trun-Pyramid joints concentrate demand near the throat and slope transitions, which can trigger rapid flank cracking once damage initiates. Overall, the FE stress/damage patterns are qualitatively consistent with the experimentally observed crack distributions and support the interpretation of the geometry-dependent performance differences reported in the subsequent sections.

3.3 Experimental studies and general discussion

In the compression-dominant tests, the nominal load-bearing footprint was $100 \times 100 \text{ mm}^2$. Jointed specimens were assembled from two cubes to a total height of 200 mm. Three fabrication routes were studied: *shotcrete*, *extrusion*, and *particle-bed* printing.

3.3.1 Experimental performance of shotcrete joint geometries

Monolithic specimens (shotcrete) The monolithic specimens showed stable compressive performance. The 90° configuration (**S1**) reached a mean peak load of 662.6 kN (i.e., $\sigma_{\max} = 66.26 \text{ MPa}$), slightly higher than the 0° counterpart (**S2**) at 638.9 kN ($\sigma_{\max} = 63.89 \text{ MPa}$). Corresponding mean strains were $\epsilon_h = 1.735$ and $\epsilon_v = 2.784$ for 90°, and $\epsilon_h = 0.870$ and $\epsilon_v = 1.932$ for 0°. Standard deviations remained modest, indicating stable performance.

The small but consistent advantage of 90° over 0° suggests that build orientation has a limited influence on compressive capacity in monolithic shotcrete; differences remain within the observed scatter. These monolithic results serve as the baseline reference for benchmarking jointed geometries and finishing strategies in the following sections.

Smooth profile joints (shotcrete) For shotcrete specimens with *Smooth* joints, the finishing route has a clear influence on the peak capacity. SAW-finished samples consistently exceeded CNC-finished ones at both orientations: **S3** (90° SAW) reached 596.6 kN ($\sigma_{\max} = 59.66 \text{ MPa}$) and **S4** (0° SAW) 583.0 kN ($\sigma_{\max} = 58.30 \text{ MPa}$), whereas **S5** (90° CNC) and **S6** (0° CNC) reached 485.3 kN ($\sigma_{\max} = 48.53 \text{ MPa}$) and 479.6 kN ($\sigma_{\max} = 47.96 \text{ MPa}$), respectively. Within each finishing method, the orientation effect remains modest (2.3% for SAW and 1.2% for CNC), indicating that interface condition and machining-induced surface state dominate over build direction for this flat contact. Mechanistically, the Smooth joint relies primarily on bearing and friction; therefore, local surface damage, edge chipping, and slight non-planarity introduced during post-processing can reduce the effective contact area and trigger earlier splitting. This interpretation is consistent with previous studies on dry concrete interfaces, where contact condition, confinement, and local stress concentration strongly govern stiffness and peak resistance [8, 9]. In this sense, the higher capacities for SAW are consistent with a less aggressive material removal process and fewer local stress raisers than multi-pass milling, which can leave small tool marks and micro-defects at the interface. Relative to the monolithic baseline, the Smooth SAW joints are roughly ~10% lower in

capacity, whereas the Smooth CNC joints are roughly $\sim 25\%$ lower, supporting the practical conclusion that SAW finishing is the more robust choice for Smooth shotcrete interfaces (Fig. 13).

Triangular profile joints (shotcrete) For shotcrete specimens with *Triangular* joints, the peak capacity reflects the combined effects of interlock geometry, build orientation, and finishing quality. At 90° , the SAW-finished specimen **S8** reached 621.5 kN ($\sigma_{\max} = 62.15$ MPa), exceeding the CNC counterpart **S7** at 533.2 kN ($\sigma_{\max} = 53.32$ MPa). At 0° , the SAW-finished specimen **S10** achieved 532.5 kN ($\sigma_{\max} = 53.25$ MPa), while the CNC-finished **S9** reached 503.6 kN ($\sigma_{\max} = 50.36$ MPa). Compared with the monolithic 90° baseline, **S8** shows a capacity reduction of about 6%, i.e., the best Triangular configuration remains close to monolithic performance. The more pronounced finishing sensitivity (SAW vs. CNC) is consistent with the presence of sharp features: small geometric imperfections and edge damage at the triangular ridges can concentrate stresses and promote early mixed-mode cracking at the notch tips. This agrees with prior studies on keyed and interlocking joints, which report strong sensitivity to corner damage, confinement, and contact quality [8–10]. Overall, these results suggest that interlocking geometry can preserve capacity, but only if the ridge quality and fit-up are controlled; otherwise, machining-induced defects and local misfit become the governing weakness.

Trun-Pyramid joint (shotcrete) The *Trun-Pyramid* specimen (**S11**, 90°) achieved a mean peak load of 591.6 kN ($\sigma_{\max} = 59.16$ MPa), which is below the monolithic reference and close to the Smooth SAW 90° level (**S3**: 596.6 kN). Although the faceted interface provides mechanical interlock and distributes compression over several bearing bands, the tapered “throat” and slope transitions still act as geometric stress concentrators, promoting localized crushing and splitting once damage initiates. This explains why the Trun-Pyramid does not outperform the best

Triangular/monolithic cases despite offering interlock: its load path is more notch-controlled, and capacity becomes governed by local damage at the transition regions rather than by global bearing.

Within the shotcrete set, the *Arc* profile provides the highest peak capacity, while the *Trun-Pyramid* specimen (**S11**) remains a strong performer among the jointed configurations. However, the Trun-Pyramid does not surpass the best Smooth–SAW or the Arc response, suggesting that its tapered load path is still governed by localized stress concentrations at geometric transitions. From a design perspective, notch- or key-type profiles (Triangular and Trun-Pyramid) are expected to become more attractive once additional actions (e.g., shear, eccentricity, or combined loading) are considered, because geometric interlock can provide kinematic restraint beyond what a flat interface offers; this aspect is noted here as a plausible implication, but it is not quantified by the present compression-only tests.

Arc (curved) joint (shotcrete) The *Arc* specimen (**S12**, 90°) attained the highest peak load among the shotcrete joints, 659.0 kN ($\sigma_{\max} = 65.90$ MPa). This is essentially on par with the monolithic references (**S1** 90° : 662.6 kN; **S2** 0° : 638.9 kN), i.e., differences are within only a few percent and should be interpreted in the context of specimen-to-specimen scatter and end-contact sensitivity. Mechanistically, the curved interface promotes a compression-dominant thrust path and spreads contact over a broader band, which reduces sharp stress concentrations at the joint plane compared with notch-controlled geometries. Consistent with this interpretation, **S12** also shows a high normalized displacement at peak ($\delta/\delta_{\max} = 0.998$; Table 3), indicating that the joint can sustain substantial deformation up to peak under the present setup. Within the available shotcrete dataset (Arc tested only at 90°), the Arc profile therefore serves as a practical upper-bound reference for compression capacity among the investigated dry-joint geometries. In selecting the joint type, manufacturing

Fig. 13 Shotcrete–Smooth joint: representative compression test sequence (Credit: iBMB TU-BS)



Table 3 Shotcrete joints-peak load (kN), gauge-averaged strains ($\%$), and normalized actuator displacement at peak for S1–S12 ($\delta_{\max} = 5.897$ mm)

ID	Joint Type	Angle	Load (kN)	ϵ_h [$\%$]	ϵ_v [$\%$]	$\delta/\delta_{\max}$ [-]
S1	Monolithic	90°	662.6	1.735	2.784	0.884
S2	Monolithic	0°	638.9	0.870	1.932	0.879
S3	Smooth (SAW)	90°	596.6	2.954	3.034	0.917
S4	Smooth (SAW)	0°	583.0	1.287	2.727	1.000
S5	Smooth (CNC)	90°	485.3	0.072	1.933	0.788
S6	Smooth (CNC)	0°	479.6	0.335	3.388	0.778
S7	Triangular (CNC)	90°	533.2	8.428	4.053	0.882
S8	Triangular (SAW)	90°	621.5	7.652	3.795	0.925
S9	Triangular (CNC)	0°	503.6	11.103	3.647	0.897
S10	Triangular (SAW)	0°	532.5	8.547	3.895	0.818
S11	Trun-Pyramid (CNC)	90°	591.6	1.309	3.073	0.950
S12	Arc (CNC)	90°	659.0	5.106	3.920	0.998

difficulty should also be considered alongside mechanical capacity.

3.4 Deformation capacity of shotcrete-printed elements

Shotcrete specimens show noticeable pre-peak deformation, but the deformation capacity is strongly geometry-dependent. Based on the actuator displacement at peak (Table 3), the SAW and Arc joints reach the largest peak displacement ($\delta/\delta_{\max} \sim 1$), followed by Trun-Pyramid ($\delta/\delta_{\max} = 0.95$), whereas Smooth–CNC peaks lower, indicating a stiffer and less tolerant response. This ranking is consistent with the expected load-path mechanics: the curved Arc interface spreads compression and delays local crushing, while a flat, fully machined Smooth interface tends to engage earlier in a more localized and less deformation-tolerant manner once local damage initiates.

Gauge-averaged vertical strains at peak remain of the same order (typically $\epsilon_v \approx 2\text{--}4\%$), whereas lateral strains ϵ_h distinguish the deformation modes. Smooth–CNC shows the smallest lateral expansion ($\epsilon_h < 0.5\%$), while Triangular joints show the largest values (about 8–11 %) due to wedge-like load redirection. Monolithic and Smooth–SAW specimens lie between these limits, whereas Arc shows moderate lateral expansion and Trun-Pyramid remains lower, consistent with their different contact engagement.

Mechanistically, Triangular joints convert part of the axial load into lateral components, Arc joints

promote a smoother compressive thrust path, and Trun-Pyramid distributes contact over multiple bearing bands.

3.5 Mechanical performance of extrusion-printed joints

Monolithic Specimens (Extrusion) Extruded monolithic specimens show a pronounced build-orientation effect in compression. The 90° configuration (E1) reaches a peak load of about 694 kN, whereas the 0° configuration (E2) peaks at about 568 kN, i.e., roughly 22% lower at 0°. This larger gap (compared with shotcrete) is consistent with extrusion's stronger layerwise fabric: loading perpendicular to the layers (90°) tends to clamp interlayer interfaces and reduce the influence of any weak bonds or voids, while loading parallel to the layers (0°) is more sensitive to interlayer weakness and facilitates splitting along preferential planes. This interpretation is consistent with the broader concrete-AM literature, where extrusion is associated with direction-dependent behaviour and layer-induced anisotropy [2, 4].

Relative to the shotcrete monolithic references (S1 at 90°, S2 at 0°), extrusion is slightly higher at 90° (on the order of a few percent) but clearly lower at 0° (on the order of 10%). Overall, the extrusion monolithic results underline that orientation must be treated as a primary variable for this process family, whereas shotcrete behaves closer to quasi-isotropic under compression.

Smooth profile joints (extrusion) For extrusion, the Smooth–CNC joint remains close to the



monolithic baseline, with a clear but not dominant orientation effect. At 90°, the peak load is about 600 kN, i.e., roughly 15% below the monolithic 90° reference, whereas at 0° it reaches about 550 kN, only a few percent below the monolithic 0° reference. The 90° case exceeds 0° by less than 10%, which is consistent with loading perpendicular to the layer interfaces and thus reducing sensitivity to interlayer weakness. In absolute terms, Smooth–CNC extrusion is comparable to shotcrete Smooth–SAW at 90° and remains higher than shotcrete Smooth–CNC, suggesting that for a flat interface the extrusion route can achieve competitive capacity provided that machining delivers good planarity and fit-up. Overall, Smooth–CNC extrusion slightly trails the monolithic benchmark but matches or surpasses the smooth variants produced by shotcrete.

Triangular joints (extrusion) For extrusion, Triangular joints are markedly finish-sensitive and perform better with SAW than with CNC. The SAW-finished specimen (E6) reaches about 465 kN, whereas the CNC-finished counterpart (E5) peaks at about 360 kN, i.e., roughly 30% lower for CNC. Even with SAW, both Triangular extrusion cases remain well below the monolithic 90° reference and also below the best shotcrete Triangular results (Fig. 14). Mechanistically, the triangular interface introduces wedge-like load paths that amplify lateral splitting and local edge stresses; any interlayer imperfections typical of extrusion can then trigger earlier cracking and localized crushing. Sawing likely produces cleaner, less-notched flanks and a more stable contact engagement, whereas multi-pass milling on steep facets can leave tool marks/micro-notches and sharper stress raisers, which is consistent with the observed CNC penalty.

Trun-Pyramid joints (extrusion) The extruded Trun-Pyramid joint (E7, CNC) reaches about 590 kN, i.e., slightly above the monolithic 0° reference and clearly above the Triangular cases. Its peak displacement is moderate (about 1.7 mm), indicating a balanced stiffness–tolerance response rather than a highly deformable behavior. This trend is consistent with the faceted geometry, which distributes compression over multiple bearing bands and reduces the extreme edge demands characteristic of wedge-type contacts. Within the extrusion set, Trun-Pyramid therefore acts as a robust “second-tier” geometry below Arc but less sensitive than Triangular to local imperfections.

Arc-shaped joints (extrusion) Across the extrusion set, the Arc joint (E8, CNC) provides the highest capacity, reaching about 645 kN. This is roughly 10%–15% above the monolithic 0° reference, yet still about 5%–10% below the monolithic 90° benchmark. The peak displacement remains relatively low (about 1.0 mm), which is consistent with a stable compressive thrust path and reduced edge stress concentrations enabled by the curved interface (Fig. 14). Notably, the extrusion Arc capacity is comparable to the shotcrete Arc, supporting the broader conclusion that curvature-driven interlock is the most reliable strategy for high compressive performance across processes.

3.6 Deformation capacity of extrusion-printed elements

In extrusion, the pre-peak deformation is comparatively limited and the *interface geometry* largely sets the deformation signature. The largest peak displacement was recorded for the *Triangular–CNC* configuration (E5, $\delta \approx 2.2$ mm, $\delta/\delta_{\max} = 1.00$), followed by the *Trun-Pyramid* (E7, $\delta/\delta_{\max} \approx 0.81$) and the *Triangular–SAW* case (E6, $\delta/\delta_{\max} \approx 0.61$). This ranking is consistent with how each profile redirects the compressive load path: Triangular ridges promote wedge-type kinematics, including local sliding/opening and earlier damage localisation; Trun-Pyramid spreads contact over multiple bearing bands; and the Arc provides a continuous thrust path that limits stress concentrations and suppresses excessive barreling.

Overall, extrusion exhibits a narrow displacement range before failure, likely due to comparatively stiff end conditions, strand-based consolidation, and efficient contact redirection in curved interfaces. Thus, even Arc joints show only modest δ despite their favourable load path.

Strain trends support this interpretation (Table 4). Vertical strains remain within a limited band (about 1.7–4.1 ‰), whereas lateral strains differentiate the response more clearly. Smooth and Triangular–CNC configurations show very small ϵ_h , while monolithic and Trun-Pyramid cases exhibit larger lateral expansion. Overall, geometry governs the deformation signature, while finishing and build orientation tune the lateral response.



Table 4 Extrusion joints-peak load (kN), gauge-averaged strains (%), and normalized actuator displacement at peak for E1–E8 ($\delta_{\max} = 2.17$ mm)

ID	Joint Type	Angle	Load (kN)	ϵ_h [%]	ϵ_v [%]	$\delta/\delta_{\max}$ [-]
E1	Monolithic	90°	693.85	4.75	3.19	0.415
E2	Monolithic	0°	568.43	1.57	2.53	0.364
E3	Smooth (CNC)	90°	595.96	0.28	1.74	0.387
E4	Smooth (CNC)	0°	554.13	1.25	3.15	0.479
E5	Triangular (CNC)	0°	363.30	0.01	2.75	1.000
E6	Triangular (SAW)	0°	464.96	0.09	2.93	0.613
E7	Trun-Pyramid (CNC)	90°	591.12	4.03	1.85	0.816
E8	Arc (CNC)	90°	645.42	1.25	4.14	0.438

Fig. 14 Experimental photo sequences: Extrusion–Arc (top), Extrusion–Triangular (middle), and particle-bed printing with a representative failure pattern from the test series (bottom). Credit: iBMB TU-BS



3.7 Mechanical performance of particle-bed-printed joints

Monolithic specimens (Particle-bed)

Particle-bed specimens show a pronounced build-orientation effect in compression, with 90° consistently outperforming 0° (e.g., **P2 (165.4 kN)** vs. **P1 (123.4 kN)**). A plausible mechanism is the process microstructure: higher porosity and weaker interlayer necks make loading *parallel* to layers (0°) more sensitive to layer separation, whereas loading *across* layers (90°) engages more effective bridging and confinement. This orientation sensitivity is stronger than in shotcrete (near-isotropic spray deposition) and typically stronger than in extrusion (better consolidation

but still layered), consistent with the predominantly brittle failure patterns observed in the particle-bed series (Fig. 14). This trend is also consistent with previous studies on particle-bed and selective-activation-based cementitious AM, where porosity and process-specific heterogeneity strongly influence mechanical response [5, 6].

Smooth joints (Particle-bed)

For particle-bed *Smooth* joints, CNC finishing provides a consistent but modest capacity gain compared with as-printed interfaces (e.g., CNC cases such as **P5 (107.2 kN)/P3 (96.5 kN)** exceed as-printed **P6 (75.1 kN)/P4 (73.7 kN)**). In absolute terms, however, these capacities remain far below the Smooth joints produced by shotcrete or

extrusion, highlighting that the governing limitation is the material microstructure (porosity and weak layer bonding) rather than surface geometry alone. CNC likely improves seating/parallelism and reduces local void-controlled loss of effective bearing area, whereas as-printed surfaces retain defects that promote premature edge crumbling; this explains the small but systematic CNC advantage and the comparatively weak orientation effect for the Smooth profile.

Triangular joints (Particle-bed)

For the *Triangular* profile, the results indicate strong sensitivity to both surface finish and build orientation. CNC finishing improves seating and reduces edge crumbling relative to the as-printed condition, while the pronounced gap between the 0° and 90° triangular cases is consistent with the weakest interlayer bonding among the three processes. The observed laminated fracture and layer separation suggest that, for particle-bed triangular interfaces, failure is governed less by geometric interlock and more by layer-parallel splitting once local stress concentrations develop (Fig. 14).

Trun-Pyramid joints (Particle-bed)

The *Trun-Pyramid* configuration benefits visibly from CNC finishing: the CNC-finished joint (**P10**, 117.5 kN) exceeds the as-printed counterpart (**P11**, 76.0 kN), placing Trun-Pyramid–CNC among the stronger particle-bed profiles. Mechanistically, this improvement is consistent with better end flatness and bearing parallelism after machining, reduced premature edge crushing, and faceted contact that

distributes compression over multiple bearing bands compared with defect-rich as-printed surfaces.

With orientation held constant, the capacity difference is therefore governed primarily by surface finish and contact quality rather than layer direction.

Arc joints (Particle-bed) For the particle-bed series, the *Arc* profile benefits markedly from CNC finishing: the CNC-finished *Arc* (**P12** (**102.9 kN**)) exceeds the as-printed counterpart (**P13** (**75.2 kN**)) by a clear margin (Table 5). Within the CNC-finished subset, *Arc*–CNC ranks below the best-performing interlocking cases (Triangular–CNC and Trun-Pyramid–CNC), but remains above all as-printed *Arc*/Smooth/Trun-Pyramid counterparts. Mechanistically, the curved interface provides a continuous bearing path and reduces stress concentrations, while machining primarily improves end flatness and seating, limiting void-driven contact losses and local crumbling. Due to printing limitations, the *Arc* cases are only available at 90° in this matrix; therefore, the observed capacity difference is governed predominantly by surface finish and contact quality rather than build orientation.

3.8 Deformation capacity of particle-bed printed elements

For particle-bed specimens, peak displacement remains within a few millimetres and is governed primarily by *surface condition* rather than by geometry-specific stiffness alone (Table 5). Across comparable profiles, as-printed interfaces generally reach larger

Table 5 Particle-bed joints–peak load (kN), gauge-averaged strains at failure (%), and normalized actuator displacement at peak for P1–P13 ($\delta_{\max} = 3.57$ mm)

ID	Joint Type	Angle	Load (kN)	ϵ_h [%]	ϵ_v [%]	$\delta/\delta_{\max}$ [-]
P1	Monolithic	0°	123.4	0.87	2.33	0.647
P2	Monolithic	90°	165.4	2.76	1.82	0.675
P3	Smooth (CNC)	0°	96.5	0.23	1.93	0.608
P4	Smooth (Print)	0°	73.7	1.74	1.84	0.947
P5	Smooth (CNC)	90°	107.2	0.45	1.33	0.641
P6	Smooth (Print)	90°	75.1	1.61	0.07	0.838
P7	Triangular (CNC)	0°	69.2	2.61	1.74	0.801
P8	Triangular (CNC)	90°	127.5	0.58	0.95	0.720
P9	Triangular (Print)	0°	79.5	0.35	0.15	0.846
P10	Trun-Pyramid (CNC)	90°	117.5	1.01	2.74	0.630
P11	Trun-Pyramid (Print)	90°	76.0	0.61	0.36	1.000
P12	Arc (CNC)	90°	102.9	1.29	3.09	0.630
P13	Arc (Print)	90°	75.2	0.34	0.18	0.762



δ at peak than CNC-finished ones (e.g., Smooth and Trun-Pyramid), indicating additional compliance from local crushing and micro-slip at rough, porous contact patches. Conversely, CNC finishing improves flatness and platen seating, reduces local rocking and edge crumbling, and therefore tends to lower the measured δ at peak despite the generally brittle material response. Arc follows the same trend: machining stabilises the continuous bearing path and limits deformation, whereas as-printed Arc exhibits higher deformation due to void-controlled contact loss and progressive crushing at the interface.

Low global deformation together with brittle or notch-like cracking indicates that stresses were concentrated at local contact imperfections rather than redistributed over the nominal bearing area. This localisation may result from rough as-printed surfaces, sharp profile features, limited bearing parallelism, particle-bed porosity, unfavourable build orientation, or non-optimal end friction. Improved finishing, rounded details, controlled platen conditions, and loading across rather than along layer interfaces can reduce these effects.

3.9 Comparison of joint geometries across fabrication techniques and orientation

3.9.1 Influence of layer orientation

Across the three processes, specimens loaded *perpendicular* to the layer planes (90°) generally achieved higher capacity than those loaded *parallel* to the layers (0°). The trend is clearest in the monolithic references and remains visible—to a lesser extent—in Smooth profiles, where geometry provides minimal stress redirection and the response is governed mainly by contact quality. Mechanistically, orientation changes how the compressive force resolves into shear along layer interfaces: loading parallel to layers increases the propensity for interlayer slip/debonding, whereas loading across layers tends to clamp interfaces and reduces interlayer shear demand. Sensitivity to orientation is strongest in particle-bed specimens (porous, necked microstructure), moderate in extrusion, and smallest in shotcrete, consistent with their relative degrees of consolidation and interface integrity.

Joint shape can partially compensate for unfavourable orientations by steering the stress path so that

interface normals align more orthogonally to layer planes, thereby lowering interlayer shear. Curved Arc joints distribute compression smoothly and reduce peak tractions; Trun-Pyramid profiles spread bearing over multiple bands; in contrast, sharp Triangular flanks act as wedges and can amplify lateral expansion and notch-driven cracking unless corners are filleted and end flatness is tightly controlled. Smooth faces are largely neutral with respect to stress redirection; their performance is therefore dominated by finishing quality (flatness, parallelism, and seating) rather than by geometry-induced confinement.

Where 90° build orientation is not feasible, selecting geometries that promote a compression-dominant load path (gentle curvature or well-angled facets), rounding sharp features, minimising tool-step scallops, and ensuring precise end finishing (sawing or CNC milling with verified parallelism) are effective measures to reduce stress concentrations and improve capacity.

3.9.2 Considering the manufacturing challenges

Because all specimens share the same nominal envelope ($100 \times 100 \text{ mm}^2$ bearing footprint), differences in machining effort can be attributed primarily to joint geometry and finishing strategy. Two robust patterns emerge from Table 2.

First, machining effort scales strongly with geometric complexity. Sawing-based finishing of simple planar profiles is fast and economical (Tri-Saw: ~ 4.6 – 4.8 m tool-path and ~ 7 – 8 min ; Smooth-Saw: $\sim 3.2 \text{ m}$ and $\sim 1 \text{ min}$), whereas CNC milling of curved or heavily faceted profiles requires substantially longer paths and cycle times (typically ~ 20 – 45 m and ~ 23 – 43 min for Smooth-Mill, Arc-Mill, and Tru-Mill). Triangular CNC sits in between, with moderate-to-high tool-paths and times comparable to Smooth-Mill.

Second, this ranking follows directly from tool-path physics and surface-quality constraints. Sawing relies on short, straight cuts and largely avoids 3D scallop-height control, so cycle time and tool wear remain low. In contrast, CNC milling of Arcs and multi-facet geometries must meet a target scallop height and maintain stable engagement over changing surface normals; this typically enforces small step-overs, multiple passes, and higher abrasive/diamond



wear, and it is more sensitive to fixture compliance and minor misalignment.

Practically, the results define a cost–performance trade-off. Tri-Saw provides a low-effort route with favourable strength-to-time balance, whereas Arc/Tru joints are justified when higher machining time is acceptable for more robust geometry-driven load transfer. For complex CNC joints, effort may be reduced by coarse roughing followed by one light finishing pass to restore fit and parallelism. The full load–deformation (P – δ) curves are provided as n Material (Online Resource 1).

Based on the framework defined in Sect. 2.5, Table 6 compares the strength-normalised joint-efficiency factors for the tested AM routes, geometries, finishing concepts, and build orientations.

Comparative interpretation. Normalising by $f_{c,comp}$ reduces route-to-route strength bias but does not remove configuration effects. For shotcrete, η_σ varies only modestly between 0° and 90° within each family (Table 6), indicating weak orientation sensitivity; the remaining differences are mainly governed by joint geometry and finishing. For extrusion, monolithic efficiencies are similar after normalisation, but some families still show directional sensitivity, especially Smooth–CNC. Particle-bed specimens show the largest spread in η_σ across orientation and surface condition, confirming that strength level alone is insufficient. Interface integrity, true bearing engagement, and process-induced anisotropy govern the

effective joint efficiency. Accordingly, σ_{max} should be treated strictly as a within-family interpolation tool.

4 Conclusion

This study examined dry-joint concrete interfaces produced by three additive-manufacturing (AM) routes—shotcrete, extrusion, and particle-bed printing—combined with robotic subtractive finishing. Rather than treating the results as isolated test outcomes, the work aimed to clarify how fabrication route, interface geometry, and finishing quality interact in governing compression-dominant dry-joint behaviour. The experimental programme comprised four joint geometries (Smooth, Triangular, Trun-Pyramid, and Arc), the available build orientations, and different post-processing strategies. In total, 120 tests were performed, including 102 compression tests on monolithic and jointed specimens and 18 batch-matched companion-cube tests, together with scan-to-CAD measurements of interface quality.

The main conclusion is that dry-joint response is governed by a coupled mechanics–manufacturing relationship rather than by nominal geometry or material strength alone. The AM route defines the material structure and defect sensitivity; the joint geometry controls force transfer and stress localisation; and the finishing method determines how effectively the designed contact surface is realised.

Table 6 Strength-normalised joint efficiency factors, $\eta_\sigma = \sigma_{max}/f_{c,comp}$, for the three AM routes ($A = 100 \times 100 \text{ mm}^2$)

Family	Shotcrete		Extrusion		Particle-bed	
	$\eta_{\sigma,90}$	$\eta_{\sigma,0}$	$\eta_{\sigma,90}$	$\eta_{\sigma,0}$	$\eta_{\sigma,90}$	$\eta_{\sigma,0}$
Monolithic	1.106	1.166	1.040	1.060	1.010	1.090
Smooth–SAW	0.996	1.064	–	–	–	–
Smooth–CNC	0.810	0.875	0.900	1.040	0.660	0.850
Smooth–Print	–	–	–	–	0.460	0.650
Tri–CNC	0.890	0.919	–	0.680	0.780	0.610
Tri–SAW	1.038	0.972	–	0.870	–	–
Trun-Pyramid–CNC	–	–	0.890	–	0.720	–
Arc–CNC	–	–	0.970	–	0.630	–

Values slightly above 1.0 may reflect systematic differences between companion cubes and joint specimens (size/boundary effects, incl. platen friction); compare mainly within the same AM route and orientation. *Additional particle-bed as-printed:* Tri–Print ($\eta_{\sigma,0} = 0.700$); Trun-Pyramid–Print ($\eta_{\sigma,90} = 0.470$); Arc–Print ($\eta_{\sigma,90} = 0.460$)

Companion strengths $f_{c,comp}$ are mean cube strengths (MPa; $n = 3$) for each route and orientation: Shotcrete ($f_{c,comp,90} = 59.9$, $f_{c,comp,0} = 54.8$), Extrusion ($f_{c,comp,90} = 66.5$, $f_{c,comp,0} = 53.5$), Particle-bed ($f_{c,comp,90} = 16.3$, $f_{c,comp,0} = 11.3$)



Reliable dry-joint behaviour therefore depends on mesostructure, surface conformity, and geometry-dependent load transfer acting together.

The three AM routes provided clearly different structural conditions for dry assembly. Shotcrete showed the highest capacity level and weakest orientation sensitivity, indicating comparatively robust compression transfer. Extrusion showed intermediate performance with clearer dependence on layer orientation. Particle-bed printing was the most sensitive route, especially to surface condition and build direction, reflecting the influence of porosity, weaker inter-layer integrity, and imperfect contact engagement.

Joint geometry also had a decisive influence. Among the investigated profiles, the Arc joint provided the most stable and consistently high compressive response, indicating that smoother redirection of the compressive force path and broader contact engagement are beneficial. Triangular joints were more sensitive to local imperfections and finishing quality, while Trun-Pyramid joints showed intermediate performance but clear stress concentration at the throat and inclined faces. Thus, joint geometry should be selected not only for nominal interlocking capacity, but also for tolerance to fabrication-induced imperfections and local contact loss.

The scan-to-CAD results and observed defects, including chipping, voids, and imperfect face conformity, support this interpretation. Such deviations reduce the true bearing area and intensify local stress concentrations. Geometric fidelity is therefore a structural requirement for reliable dry-joint behaviour, not only a manufacturing-quality issue.

From a practical perspective, mechanical performance must be balanced against fabrication effort. CNC milling improved geometric control for complex profiles but required longer machining time. SAW finishing offered a simpler route for planar or sharp-edged interfaces and, in several cases, achieved equal or higher compressive capacity than CNC for nominally simple joints. Accordingly, Arc-CNC is the most robust option when mechanical reliability is prioritised, whereas Triangular-SAW provides a favourable performance-effort balance when production efficiency is decisive.

The findings provide comparative guidance within the investigated specimen size, materials, and boundary conditions. The main limitations are the incomplete orientation matrix for some geometry-route

combinations and route-dependent material differences. Future work should address shear, tension, cyclic and combined loading, complete the missing orientation cases, and develop design-oriented approaches integrating geometry, contact quality, and manufacturability for segmented AM concrete construction.

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Data availability The datasets generated and analysed during the current study are available from the corresponding author upon reasonable request. Upon acceptance, the full curated dataset will be deposited in the institutional repository of TU Braunschweig.

Declarations

Conflict of interest The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analysis, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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